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Concentration Inequalities (AI2200)

Lecture 02

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Cramér—Chernoff Bound

Theorem 1 (Chernoff Bound)

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Let X be a random variable such that for some fixed $\varepsilon > 0$, the MGF of X satisfies

$$M_X(t) < +\infty \quad \text{for all } t \in (-\varepsilon, \varepsilon)$$

Then, for any $\alpha \in \mathbb{R}$,

$$\mathbb{P}(\{X \geq \alpha\}) \leq \frac{M_X(t)}{e^{t\alpha}} \quad \forall t > 0.$$

Optimising the RHS over $t > 0$, we get

$$\mathbb{P}(\{X \geq \alpha\}) \leq \inf_{t>0} \frac{M_X(t)}{e^{t\alpha}}, \quad \alpha \in \mathbb{R}.$$

Cramér—Chernoff Bound

Proof of Theorem 1:

- Keeping α fixed, notice that for any $t > 0$, we have

$$\begin{aligned}\mathbb{P}(\{X \geq \alpha\}) &= \mathbb{P}(\{tX \geq t\alpha\}) \\ &= \mathbb{P}(\{e^{tX} \geq e^{t\alpha}\}) \\ &\leq \frac{\mathbb{E}[e^{tX}]}{e^{t\alpha}} \quad (\text{Markov's inequality applied to } e^{tX}).\end{aligned}$$

- Because the RHS holds for any $t > 0$, we can optimise over $t > 0$ to get the desired result

Cramér–Chernoff Bound

- $\mathbb{P}(\{X \geq \alpha\})$ is typically called **upper tail** probability
- One may also derive upper bounds on the **lower tail** probabilities: for any fixed $\alpha \in \mathbb{R}$ and for all $t < 0$,

$$\begin{aligned}\mathbb{P}(\{X \leq \alpha\}) &= \mathbb{P}(\{tX \geq t\alpha\}) \\ &= \mathbb{P}(\{e^{tX} \geq e^{t\alpha}\}) \\ &\leq \frac{M_X(t)}{e^{t\alpha}}.\end{aligned}$$

- Because the above relation holds true for all $t < 0$, we may optimise over $t < 0$ to get

$$\mathbb{P}(\{X \leq \alpha\}) \leq \inf_{t < 0} \frac{M_X(t)}{e^{t\alpha}}, \quad \alpha \in \mathbb{R}.$$

Cramér–Chernoff Upper Tail Bound

$$\mathbb{P}(\{X \geq \alpha\}) \leq \inf_{t > 0} \frac{M_X(t)}{e^{t\alpha}}, \quad \alpha \in \mathbb{R}.$$

Cramér–Chernoff Lower Tail Bound

$$\mathbb{P}(\{X \leq \alpha\}) \leq \inf_{t < 0} \frac{M_X(t)}{e^{t\alpha}}, \quad \alpha \in \mathbb{R}.$$

Example: Pareto Distribution

- Suppose that X is a continuous random variable with PDF

$$f_X(x) = \frac{3}{x^4}, \quad x \geq 1 \quad \text{(Pareto distribution)}.$$

- Then, MGF of X is given by

$$M_X(t) = \mathbb{E}[e^{tX_1}] = 3 \int_1^\infty \frac{e^{tx}}{x^4} dx = \begin{cases} +\infty, & t > 0, \\ \leq 1, & t \leq 0. \end{cases}$$

- Applying Theorem 1, we get that

$$\mathbb{P}(\{X \leq \alpha\}) \leq \inf_{t < 0} \frac{M_X(t)}{e^{t\alpha}} = \exp \left(\inf_{t < 0} \{ \log M_X(t) - t\alpha \} \right).$$

Sums and Averages of IID Random Variables

- Suppose $X_1, X_2, \dots$ are **independent and identically distributed (IID)** with mean $\mu \in \mathbb{R}$
- Given $n \in \mathbb{N}$, let

$$S_n := \sum_{\ell=1}^n X_{\ell}.$$

- From the strong law of large numbers, we know that

$$\frac{S_n}{n} \xrightarrow[n \rightarrow \infty]{\text{a.s.}} \mu.$$

Concentration of $\frac{S_n}{n}$

How well does $\frac{S_n}{n}$ concentrate around its mean μ ?

Concentration of $\frac{S_n}{n}$ Around its Mean

- We have

$$M_{\frac{S_n}{n}}(t) = \left(M_{X_1} \left(\frac{t}{n} \right) \right)^n, \quad t \in \mathbb{R}.$$

- By Theorem 1, we get that for any $\varepsilon > 0$,

$$\mathbb{P} \left(\left\{ \frac{S_n}{n} \geq \mu + \varepsilon \right\} \right) \leq \inf_{t > 0} \frac{\left(M_{X_1} \left(\frac{t}{n} \right) \right)^n}{e^{t(\mu + \varepsilon)}}, \quad \mathbb{P} \left(\left\{ \frac{S_n}{n} \leq \mu - \varepsilon \right\} \right) \leq \inf_{t < 0} \frac{\left(M_{X_1} \left(\frac{t}{n} \right) \right)^n}{e^{t(\mu - \varepsilon)}}.$$

Example: Bernoulli Distribution

- Suppose $X_1, X_2, \dots \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$
- Then, we have

$$M_{X_1}(t) = \mathbb{E}[e^{tX_1}] = (1-p)e^0 + pe^t = 1-p+pe^t, \quad t \in \mathbb{R}.$$

- If $S_n = \sum_{\ell=1}^n X_\ell$ for $n \in \mathbb{N}$, then

$$M_{S_n}(t) = \mathbb{E}[e^{tS_n}] = \mathbb{E}[e^{t \sum_{\ell=1}^n X_\ell}] = (1-p+pe^t)^n, \quad t \in \mathbb{R}.$$

Markov's Inequality

Objective

Suppose $X_1, X_2, \dots \stackrel{\text{iid}}{\sim} \text{Ber}(0.5)$ and $S_n = \sum_{\ell=1}^n X_\ell$. Upper bound $\mathbb{P}(\{S_n \geq 3n/4\})$.

- **Markov's inequality:**

$$\begin{aligned}\mathbb{P}\left(\left\{S_n \geq \frac{3n}{4}\right\}\right) &\leq \frac{\mathbb{E}[S_n]}{3n/4} \\ &= \frac{n/2}{3n/4} \\ &= \frac{2}{3}.\end{aligned}$$

Chebyshev's Inequality

Objective

Suppose $X_1, X_2, \dots \stackrel{\text{iid}}{\sim} \text{Ber}(0.5)$ and $S_n = \sum_{\ell=1}^n X_\ell$. Upper bound $\mathbb{P}(\{S_n \geq 3n/4\})$.

- Chebyshev's inequality:**

$$\begin{aligned} \mathbb{P}\left(\left\{S_n \geq \frac{3n}{4}\right\}\right) &= \mathbb{P}\left(\left\{S_n - \mathbb{E}[S_n] \geq \frac{3n}{4} - \mathbb{E}[S_n]\right\}\right) \\ &= \mathbb{P}\left(\left\{S_n - \frac{n}{2} \geq \frac{n}{4}\right\}\right) \\ &\leq \mathbb{P}\left(\left\{\left|S_n - \frac{n}{2}\right| \geq \frac{n}{4}\right\}\right) \\ &\leq \frac{\text{Var}(S_n)}{(n/4)^2} \\ &= \frac{n/4}{n^2/16} = \frac{4}{n}. \end{aligned}$$

Chernoff's Bound

Objective

Suppose $X_1, X_2, \dots \stackrel{\text{iid}}{\sim} \text{Ber}(0.5)$ and $S_n = \sum_{\ell=1}^n X_\ell$. Upper bound $\mathbb{P}(\{S_n \geq 3n/4\})$.

- Chernoff's Bound:**

$$\begin{aligned}
 \mathbb{P}\left(\left\{S_n \geq \frac{3n}{4}\right\}\right) &\leq \inf_{t>0} \frac{M_{S_n}(t)}{e^{3nt/4}} \\
 &= \inf_{t>0} \frac{(0.5 + 0.5e^t)^n}{e^{3nt/4}} \\
 &= \inf_{t>0} \exp\left(n \log(0.5 + 0.5e^t) - \frac{3nt}{4}\right) \\
 &= \exp\left(\inf_{t>0} \left\{n \log(0.5 + 0.5e^t) - \frac{3nt}{4}\right\}\right) \\
 &= \left(\frac{16}{27}\right)^{n/4}.
 \end{aligned}$$

Comparison of the Three Upper Bounds

Setup

$$X_1, X_2, \dots \stackrel{\text{iid}}{\sim} \text{Bernoulli}(0.5), \quad S_n = \sum_{\ell=1}^n X_\ell.$$

Inequality	Quantity used	Bound on $\mathbb{P}(\{S_n \geq 3n/4\})$	Decay with n
Markov	$\mathbb{E}[S_n] = n/2$	$\frac{2}{3}$	Constant
Chebyshev	$\text{Var}(S_n) = n/4$	$\frac{4}{n}$	Polynomial, $O(n^{-1})$
Chernoff	$M_{S_n}(t) = \left(\frac{1+e^t}{2}\right)^n$	$\left(\frac{16}{27}\right)^{n/4}$	Exponential

Chernoff's bound captures the exponential decay of tail probability $\mathbb{P}(\{S_n \geq 3n/4\})$

Concentration of Bernoulli Averages

Theorem 2

Suppose $X_1, X_2, \dots \stackrel{\text{iid}}{\sim} \text{Ber}(p)$ for some fixed $p \in (0, 1)$. Then, the following hold.

- Upper tail probabilities:

$$\mathbb{P} \left(\left\{ \frac{S_n}{n} - p \geq \varepsilon \right\} \right) \begin{cases} = 0, & \varepsilon \geq 1 - p, \\ \exp(-n D_{\text{KL}}(\text{Ber}(p + \varepsilon) \parallel \text{Ber}(p))), & 0 < \varepsilon < 1 - p. \end{cases}$$

- Lower tail probabilities:

$$\mathbb{P} \left(\left\{ \frac{S_n}{n} - p \leq -\varepsilon \right\} \right) \begin{cases} = 0, & \varepsilon \geq p, \\ \exp(-n D_{\text{KL}}(\text{Ber}(1 - p + \varepsilon) \parallel \text{Ber}(1 - p))), & 0 < \varepsilon < p. \end{cases}$$

- Proof is left as **exercise**